MILITARY SPECIFICATION

TRANSMITTER, IONOSPHERIC SOUNDING, OBLIQUE INCIDENCE (AN/FPT-11)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Purpose. This specification covers the performance requirements for an Oblique Incidence Ionospheric Sounder transmitter. The transmitter automatically transmits binary phase coded signals of short duration on 160 discrete frequencies over the 2-32 megaHertz (MHz) range for use in determining optimum frequencies for high frequency (HF) communications. This equipment is for use in conjunction with sounder receiving equipment aboard naval ships and at shore communications stations.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-H-46855	- Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-M-38510	- Microcircuits, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

FSC 5840

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MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-242	- Electronic Equipment Parts (Selected Standards)
MIL-STD-415	- Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	- Maintainability/Verification/ Demonstration/Evaluation
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-810	- Environmental Test Methods
MIL-STD-1345	- Data Measurement, in Support of Maintenance, Calibration and Repair of Electronic Equipment
MIL-STD-1364	- Standard General Purpose Electronic Test Equipment
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1631	- Procedure for Selection of Electronic and Electrical Parts and Materials During Design of Military Items

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3. The sample(s) shall be suitable for complete evaluation of mechanical and electrical form, design, and performance. The sample(s) shall employ standard and approved non-standard parts and be completely representative of final equipment.

3.2 Parts selection. Standard parts as specified in MIL-STD-242 and approved nonstandard parts shall be used. Class B devices in accordance with MIL-M-38510 shall normally be used in all applications of this specification.

3.2.1 Nonstandard parts. Action on approval of nonstandard parts shall be in accordance with Requirement 22 of MIL-STD-454. Certification of first article sample tests or acceptance for delivery shall not be made by the cognizant Government Representative for any equipment using unapproved nonstandard

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parts without the written consent of the command or agency concerned. Prescreening of nonstandard parts approval requests shall be accomplished in accordance with MIL-STD-1631.

3.3 Temperature range. Operating and nonoperating temperature ranges shall conform to Range 4 equipment in accordance with MIL-E-16400.

3.4 General requirements. The equipment shall be in accordance with the provisions of MIL-E-16400 except as hereinafter specified. Tests for inclination, airborne noise, structural integrity, and magnetic properties are excepted.

3.4.1 Operational requirements. This transmitter shall give reliable performance within the limits specified hereinafter under conditions of 24 hours per day continuous operation at a 6 percent maximum duty cycle during exposure to the conditions normally encountered in Naval Shore Communications service. This transmitter shall be capable of providing rated radio frequency (RF) output power of 30 kilowatts (kw) effective envelope pulse power as determined by the method specified in 3.5.1.5.2 on each and every transmitted frequency over the specified frequency range, and shall not deviate more than 1 decibel (dB) below or 2 dB above the 30 kw level. The transmitter shall be capable of operating into an omnidirectional antenna, through a 50 ohm unbalanced transmission line, with a voltage standing wave ratio (VSWR) of 3 to 1 or less, as specified in 3.5.1.5.3.

3.4.1.1 Primary power source. The transmitter amplifier group and power unit shall operate from a 200, 208, 220, 230 or 240 volt (V), 50/60 hertz (Hz) ±5 percent, 3 phase power source. The exciter frequency generating unit shall operate from a 115/230 ±10 percent, 50/60 Hz single phase source. Input power to the transmitter shall be connected through a suitable rated, built-in circuit breaker.

3.4.1.1.1 Step start. Step start circuitry shall be employed in the power amplifier high voltage power supply to limit the prime power inrush current. The maximum stepping time shall be 50 milliseconds (ms).

3.4.2 Operating conditions. The equipment shall conform to the requirements of this specification under the following conditions:

(a) Ambient temperature: 0° to +50°C operating
(b) Ambient humidity: up to 95 percent relative humidity
(c) Electromagnetic field: RF field power ambient of 10 V per meter
(d) Elevation: sea level to 10,000 feet
(e) Availability: a minimum of 99.6 percent of the time

3.4.3 Shock and vibration. The equipment shall be designed and constructed to withstand the shock and vibration encountered during transportation and handling when subjected to the rough handling test of MIL-P-116 and the vibration test method 514.1, procedure X, Curve AY, of MIL-STD-810.

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3.4.4 Thermal survey. A thermal survey is required.

3.4.5 Reliability. The equipment covered by this specification shall have a mean-time-between-failure (MTBF), θ_o, of 1500 hours.

3.4.6 Maintainability.

3.4.6.1 Maintainability requirements.

3.4.6.1.1 Quantitative maintainability requirements. The equipment shall have an equipment repair time (ERT) of no greater than 0.75 hours when repair is accomplished by module and replaceable parts. The ERT includes localization, isolation, disassembly, interchange, reassembly, alignment and checkout for all modules and parts.

3.4.6.1.2 Qualitative maintainability requirements. The equipment shall be constructed to provide ease of maintenance, accessibility and replacement of modules and replaceable parts.

3.4.6.1.2.1 Plug-in construction. Optimum use shall be made of plug-in construction.

3.4.6.1.2.2 Interchangeability. Unless otherwise approved by the command or agency concerned, mechanical and electrical interchangeability shall exist between similar assemblies, subassemblies, and replaceable parts supplied by a contractor. In the design of the equipment, provisions shall be made for design tolerance to accommodate various articles such as microcircuits, tubes, resistors, and other parts having the limiting dimensions and characteristics set forth in the specifications for the particular part involved without departure from the specified performance.

3.4.6.2 Instrumentation requirements.

3.4.6.2.1 Circuit test points. Circuit test point accesses shall be provided which comply with the criteria outlined in MIL-STD-415. The following types of tests shall be considered for determining the test points: (1) signal quality, (2) circuit performance, (3) calibration, (4) system alignment, (5) installation and checkout, (6) planned maintenance systems tests, (7) line replaceable unit (LRU) fault location, and (8) shop replaceable part (SRP) fault isolation. Test points for calibration shall be provided external to unit under test (UUT), except where it would affect the calibration. In addition, test points shall be provided to the LRU interconnection connector for fault isolation of SRP's (non-repairables) or pertinent functional circuits within the LRU whichever is the most desirable to fault isolate (test) by shop programmable test instrumentation.

3.4.6.2.2 Test equipment built-in. A five inch (127 mm) diameter cathode ray oscilloscope (CRO) shall be mounted within the transmitter and placed at the proper height for ease of use and viewing. Vertical and horizontal sweep speeds and sensitivity shall be compatible with any wave form generated in the

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transmitter. The CRO shall be capable of functioning as a normal oscilloscope for maintenance and trouble shooting, and for displaying wave forms from switch selected hardwired inputs. Hardwired inputs shall be provided for: External time signals for synchronizing the transmitter; RF signal outputs of the exciter, driver, power amplifier; and those digital signals necessary to verify the proper operation of the transmitter. Provision shall be made for automatic synchronization of the CRO for the waveform being displayed.

3.4.6.2.3 Meters. Meters shall be provided to indicate all critical circuit parameters, namely: Plate voltages and currents, grid bias and screen voltages, and exciter and power amplifier output power. One meter in the exciter and power amplifier groups shall be capable of being switched to any circuit requiring periodic checking or adjustment. All meters shall have an accuracy within ±5 percent.

3.4.6.2.3.1 Elapsed time meter. An elapsed time meter shall be provided in the equipment to indicate operating hours defined as filament voltage on-time.

3.4.6.2.4 Labeling. All meters, test points, and operating service controls shall be labeled for ease of identification and use. Chassis covers and plug-in assemblies shall be provided with a stenciled parts layout to assist in locating parts during maintenance and repair.

3.4.6.3 Support test instrumentation.

3.4.6.3.1 Shop maintenance, repair and calibration tests. LRU's shall be designed to make SRP (non-repairable) fault isolation test points available for interconnection to shop programmable test equipment. The test leads should be included in the interconnection connector except where cross talk would upset the measurement, then it may be brought out to a connector separately.

3.4.7 Human engineering.

3.4.7.1 Human factors criteria. In order to achieve more effective man/machine integration and utilization from the standpoint of both operation and maintenance, the contractor shall comply with the Human Factors Design Criteria of MIL-STD-1472 and the requirements specified herein. If off-the-shelf equipment (see 6.5) is used, it shall be selected as required by the Standardization paragraph of MIL-STD-1472.

3.4.7.2 Human engineering aspects of detail design. Equipment design shall conform to the criteria of MIL-STD-1472 and the requirements of this specification.

3.4.7.2.1 Controls, indicators and panel layouts. The design of operator and maintainer panel layouts and controls shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, Indicator and Panel Layout paragraph of MIL-E-16400, except that color coding shall be in accordance with MIL-STD-1472. The commencement of the first article testing prior to

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the approval of the submittal requirements of MIL-E-16400 shall be at the contractor's risk. Tactile identification and dark adaptation requirements are not applicable to this equipment.

3.4.7.2.2 Alarm lamps. Lamps used on front panels for alarms and other purposes shall conform to the requirements of the Transilluminated Displays paragraph of MIL-STD-1472. A capability for remote indication of the status of the alarm lamps shall be provided. Legend lights shall be used in preference to simple indicator lights except where design considerations demand that simple indicators be used. Lamps used in legend switches shall conform to the requirements of the Legend Switches paragraph of MIL-STD-1472.

3.4.7.2.3 Auditory alarms. The requirements of the Auditory Displays paragraph of MIL-STD-1472 shall be followed in the design of auditory alarms. When a single tone audible alarm is provided, the alarm shall be of the summary alarm type which has a shutoff switch with an automatic reset feature. The audible alarm shall have a lamp which is not affected by the shutoff switch. The audible alarm, shutoff switch, and the lamp shall be enclosed by a line on the front panel, labeled SUMMARY ALARM, and be connected to other alarm lamps in the system. If more than one single tone audible alarm is provided, the requirements of this paragraph shall apply except that each alarm shall have a different, distinguishable tone and each alarm shall be labeled SUMMARY, ALARM, _____ EQUIPMENT. Indication of the audible alarm status shall also be displayed remotely.

3.4.7.3.4 Safety. The equipment design shall promote maximum safety for both operational and maintenance personnel during all phases of operational life in accordance with Requirement 1 of MIL-STD-454.

3.4.7.3.5 Acoustic noise. The transmitter shall produce no more than 60 dB of acoustic noise measured by instrumentation having 70 dB weighting characteristic (B weighting, ASA S1.4 - 1961) at a distance of 10 feet (300 cm) in any direction normal to the nearest outer surface of the transmitter when it is installed and operated in a large transmitter room without special acoustic treatment, and with an ambient noise level of not greater than 50 dB. (Reference noise level is a free progressive 1000 Hz sound wave of 10-16 watts/cm^2).

3.4.8 Maintenance concept. The maintenance concept for this equipment shall provide for its maximum utilization for its intended purpose by the operational forces. To support this concept, this equipment shall be modularly constructed, with modules (sub-assemblies) that can be rapidly fault isolated, removed, and replaced by the operational forces at the organizational level. Modules/printed circuit boards will not normally be piece part repaired at the organizational level unless a level of repair (LOR) analysis shows that this is the most economical maintenance support posture or is dictated by operational readiness requirements. In addition to consideration for repairing modules for reuse (return to service), the discard at failure alternative will be utilized when determined by LOR analysis to be the economical maintenance action. Design for discard is encouraged and discard screening procedures are required in support of the discard mode to minimize false discard rates.

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3.4.8.1 Organizational maintenance. Organizational maintenance of this equipment shall consist of minor inspection, routine cleaning and servicing, lubrication, adjustments, and the removal and replacement of assemblies, modules or chassis mounted components determined to be faulty. Operational tests will be accomplished to verify proper operation and performance. All preventive maintenance will be performed at the organizational maintenance.

3.4.8.2 Intermediate maintenance. Intermediate maintenance of this equipment shall be within the capabilities of the using organization. Intermediate maintenance secondary support consists of piece part fault isolation, removal, replacement, and checkout of modules/printed circuit cards removed for repair, as primary maintenance actions by the organizational maintenance level. It includes calibrating, aligning, manufacture of critical non-available parts (mechanical), and performance verification testing to ensure proper unit operation. Maintenance actions assigned for this level of maintenance must be supported by a level of repair analysis or be dictated by operational readiness requirements.

3.4.8.3 Depot maintenance. Depot maintenance of this equipment shall be within the capabilities of the maintenance facilities of the logistic organizations of the three military services. Depot level maintenance will normally consist of inspection, test, repair, modification, alteration, modernization, conversion, overhaul, reclamation, or rebuild or parts, assemblies, sub-assemblies, components, equipment end items and weapon systems and manufacture of critical non-available parts. Depot level maintenance shall also include fault isolation of assemblies/sub-assemblies to the component level, removal/replacement of faulty components, and verification testing to ensure proper performance. Maintenance actions assigned to this level of maintenance must be supported by a level of repair analysis or be dictated by operational readiness/capability considerations.

3.4.8.4 Skill levels. Skill levels at the 3 maintenance levels are as follows:

(a) Organizational - ET-3 and ETSN (E-4 and E-3)
(b) Intermediate - ETC, ET-1 and ET-2 E-7, E-6 and E-5)
(c) Depot - Electronic mechanic

3.4.9 Electromagnetic compatibility.

3.4.9.1 Compatibility. The transmitter shall meet the requirements of MIL-STD-461 for class 1B equipments except as follows:

3.4.9.2 Radiated susceptibility. The field intensity for use in test RS03 shall be 10 V per meter.

3.4.9.3 Spurious emission. The power level of all spurious emissions, including harmonics, distortion, parasitic oscillations and intermodulation products outside of the normally transmitted bandwidth shall be as follows: Any emission appearing on any frequency removed from the carrier frequency by at least 50 percent, but not more than 100 percent of the maximum transmitted

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bandwidth, shall be attenuated not less than 50 dB below the carrier level. Any spurious or harmonic emission appearing on any frequency removed from the carrier frequency by at least 100 percent of the maximum transmitted bandwidth shall be attenuated below the carrier level by not less than 60 dB.

3.5 Detailed performance.

3.5.1 Electrical.

3.5.1.1 Frequency range. The frequency range shall be 2.0 to 32.0 MHz. The transmitter shall be capable of transmitting 160 and 80 discrete frequencies spaced in linear progression as follows:

For 160 discrete frequencies:	For 80 discrete frequencies:
2.025 to 3.975 MHz in 50 kHz steps	2.075 to 3.975 MHz in 100 kHz steps
4.050 to 7.950 MHz in 100 kHz steps	4.150 to 7.950 MHz in 200 kHz steps
8.100 to 15.900 MHz in 200 kHz steps	8.300 to 15.900 MHz in 400 kHz steps
16.200 to 31.800 MHz in 400 kHz steps	16.600 to 31.800 MHz in 800 kHz steps

3.5.1.2 Transmission cycle. Frequencies shall be transmitted in pulse form and stepped in ascending order. Stepping through the specified frequency range shall be initiated by a clock programmer to a precision of within 1 ms. The programmer shall select, order, and control emission from the transmitting equipment on the frequency, in the sequence, and to the timing accuracies specified herein. Initiation of the transmission cycle shall be programmable in increments of 100 ms within a one minute period. The transmission cycle shall be 16 seconds in length and automatically repeated every minute; even though it shall be possible for the selection or deletion of from 1 to 159 frequencies to be transmitted.

3.5.1.3 Pulses per frequency. There shall be two pulses per frequency spaced 50 ms apart per transmission cycle. Pulses between adjacent frequencies shall be 50 ms with 160 discrete frequencies and 150 ms when operating on 80 discrete frequencies.

3.5.1.3.1 Pulse repetition frequency (PRF). The system shall operate at a specific repetition rate of 20 pulses per second (PPS) within the accuracy derived from the frequency standard of 3.5.1.4.

3.5.1.3.2 Pulse characteristics. The transmitter output shall be 13 phase modulated subpulses within a 2.6 ms time base. Each of the 13 subpulses shall be shaped to provide maximum energy in a bandwidth not to exceed 7.5 kilohertz (kHz) between 3 dB points. Modulation form of the subpulses shall be binary phase coding of 0 and 180 degrees in a 13 bit Barker code which has the best autocorrelation function with a side lobe level of a single unit as compared to 13 for the main pulse. The code is 1010110011111 where 0 and 1 denotes a phase shift of 180 degrees.

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3.5.1.3.2.1 Subpulse shape. Each of the individual subpulses comprising the transmitted pulse shall be of symmetrical trapezoidal form with gaussian shaped leading and falling edges having rise and fall times of 20 microseconds (µs) ±5 percent as measured from the 10 percent to 90 percent amplitude points.

3.5.1.4 Time and frequency standard.

3.5.1.4.1 Time and frequency accuracy. The transmitting equipment shall include a master oscillator (internal frequency standard) from which all timing and frequency generation functions within the equipment shall be derived. After initial (or subsequent) warm-up and correction, the basic output (or outputs) from this standard shall not vary more than one part in 10^{10} per day. With monthly correction, the output(s) shall not vary more than three parts in 10^{9} per year. Time derived from the standard shall be displayed on the front of the transmitter in terms of hours, minutes, and seconds.

3.5.1.4.1.1 External frequency standard. A shielded radio frequency (rf) connector or connectors shall be provided for connecting the rf output from an external frequency standard through a double shielded RG-58/U coaxial cable as a substitution for the interfial frequency standard within the transmitter.

3.5.1.4.2 Transmitter output signal phase stability. The phase stability of the transmitted signals shall not exceed a rate of change greater than 0.044 radian in 10 ms.

3.5.1.4.2.1 Changes in the primary power supply voltage of ±10 percent or ambient temperature changes of 20°C over the range 0° to +50°C shall not cause the basic output (or outputs) of the internal standard to vary more than one part in 10^{10} from nominal value.

3.5.1.4.3 Transmitter programmer. The programmer shall select, order, and control emission of the transmitter on prescribed frequencies. The programmer shall control emission timing of the transmitter with respect to the system sequential timing cycle and shall control the transmission or deletion of any frequency or combination of frequencies from the transmission cycle.

3.5.1.4.4 Shielded input. A shielded input connector shall be incorporated on the front panel to provide for phase comparison and correction of the master oscillator with 1 MHz or 100 kHz primary standard signals of 1 V across 50 ohms. Programmer timing correction shall be accomplished by synchronization with very low frequency (VLF) or HF standard time signals, to an accuracy of at least ±1 ms.

3.5.1.4.5 Battery-power supply. The master oscillator and the time deriving circuitry required to maintain synchronism with a standard time and frequency broadcast shall be provided with a floating battery power supply which will automatically operate in the event of failure of normal primary power. The battery supply shall have a capacity of not less than six hours operating time before system degradation from the established requirements specified herein.

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Circuitry shall be provided for automatically charging the battery. A meter shall be provided to monitor the state of battery charge. The battery shall conform to Requirement 27 of MIL-STD-454.

3.5.1.5 RF amplifier and output circuitry.

3.5.1.5.1 Output impedance. Output impedance shall be a nominal 50 ohms, unbalanced.

3.5.1.5.2 Power output. The effective envelope pulse power output of the transmitter shall be 30 kw based on a peak to peak voltage measurement of the pulse envelope when the transmitter is delivering power into a 50 ohm resistive load to six percent maximum duty cycle and at 20 pulses per second. The output power (P_O) shall be computed from the formula:

$$P_O = \frac{\left(\frac{\text{PEAK-TO-PEAK ENVELOPE VOLTAGE}}{2\sqrt{2}}\right)^2}{50 \times 1000} \text{ KILOWATTS}$$

3.5.1.5.3 VSWR matching capabilities. The transmitter shall be capable of delivering full rated power output within the limits of +2 dB and -2.5 dB into a 50 ohm transmission line which may have a load impedance consisting of any combination of R ±jX resulting in values of VSWR up to 3 to 1. A protection device shall be included which will remove power from the final amplifier prior to the occurance of damage if the VSWR at any time should exceed 5:1.

3.5.1.5.4 Output power metering. In addition to any other means, power output shall be measured by means of meter indicators provided to simultaneously display true forward and reverse peak effective power in kilowatts. Accuracy of indicators shall be within ±5 percent.

3.5.1.5.5 High VSWR and crowbar protection circuits. High VSWR (5 to 1) and electronic Crowbar protection circuits shall be provided for the power amplifier and associated power supply. Spark gaps shall be provided in appropriate locations in all high voltage and RF power circuits. VSWR sensing devices shall incorporate means for adjustment of operating points over a VSWR range from 3:1 to 5:1. Visual and audible alarms shall be provided to indicate operation of protection circuits. The status of the alarms shall also be displayed in appropriate remote locations.

3.5.1.5.6 Circuitry. Suitable circuitry shall be incorporated within the transmitter for the testing of all VSWR and electronic Crowbar protection circuits.

3.5.1.5.7 Power protection. Protection shall be positive and operate at speeds suitable to the preservation of the tubes and circuitry in the transmitter. Manual controls shall be provided to return the transmitter to operation.

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3.5.1.5.8 Power output control (manual). A means shall be provided to reduce power by 6 dB in 1 dB steps. The power output at any given step shall be maintained within -1 dB or +2 dB.

3.5.1.5.9 Manual operation. The transmitter shall have the capability of being easily set manually to any desired frequency referenced in 3.5.1.1. The transmitter shall be capable of full power output (30 kw) PEP while stopped on any selected frequency. A visual, front panel indication shall be provided to indicate the particular frequency in use.

3.5.1.5.10 RF power amplifier and driver tubes. Selection of RF power amplifier and driver tubes shall not abrogate the reliability requirement of this specification.

3.5.1.5.11 Filament voltage regulation. Filament voltage for the power amplifier and driver tubes shall be regulated within the voltage tolerance specified by the tube manufacturer for the type of service required herein.

3.5.2 Mechanical.

3.5.2.1 Panel layout and arrangement of operating controls. The panel layouts shall be such as to allow easy use of all controls and indicators.

3.5.2.2 RF output connector. The transmitter RF output fitting shall be of a type suitable for the connection of RG-321 or equivalent gas or air filled transmission line. The connector impedance shall be a nominal 50 ohms.

3.5.2.3 Enclosures. The equipment shall be mounted in a cabinet group of not more than three cabinets having overall combined dimensions not exceeding 86 inches (2184.4 mm) high by 72 inches (1828.8 mm) wide by 30 inches (762 mm) deep while maintaining specified performance and ease of maintenance and repair.

3.5.2.3.1 The degree of enclosure shall be totally enclosed in accordance with MIL-STD-108 except for openings required for forced air cooling, control shafts, and cable connectors.

3.5.2.3.2 Accessibility. The arrangement of all parts shall be such that replacement or adjustment of any part is possible without removal of or damage to adjacent parts. All parts including power supply components, shall be readily accessible for replacement or repair. All parts, with chassis withdrawn or access doors open, shall be accessible from the front of the enclosure.

3.5.2.3.3 Slide-tracks. Tracks, except for heavy or bulky component mountings, shall be provided on all chassis mountings to facilitate withdrawal of chassis for maintenance. The enclosure and slide construction shall permit each chassis to be fully withdrawn from the enclosure and supported by the slides in the withdrawn position. In addition, and depending upon chassis location in the enclosure, it shall be possible to rotate each chassis while in the withdrawn position and to lock it in five positions; namely, vertical (front panel up), vertical (front panel down), horizontal and two 45 degree positions in between the first three positions.

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3.5.2.3.4 All enclosure cabling to each chassis shall be of sufficient length to permit operation of the equipment when the chassis is withdrawn from the enclosure on slides. All plug-in subassemblies shall be operable by use of test cables or plug-in extenders. Test cables and plug-in extenders shall be provided with each equipment.

3.5.2.3.5 Equipment cooling. The enclosure shall be designed for adequate forced ambient air cooling of the equipment, including provision for venting of exhaust air to an external plenum chamber. Individual chassis shall contain, where necessary, independent blowers and shall exhaust air in such a manner that it does not recirculate back into the chassis from which it was exhausted or into any other chassis of this equipment.

3.5.2.3.6 External connections. All external connections shall have provision for entry at either top or bottom of the back of the enclosure except as required in 3.5.1.4.4.

3.5.2.3.7 Interlocks. Each chassis and access door shall be provided with an electrical interlock switch with an integral bypass device. This bypass device shall be manually operable, and shall be such that returning the chassis to the enclosure will automatically open the manual bypass and reset the interlock to function normally. Mechanical interlock systems such as those employing cams, levers, rods, and so forth, shall not be used. All interlock systems shall be provided to protect personnel and prevent damage to equipment during normal operation or while performing maintenance or repair.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include: (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

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4.1.2 Support maintenance test procedure verification.

4.1.2.1 Test equipment selection. Test equipment used to verify the support maintenance test procedures shall be selected from MIL-STD-1364. Test equipment not specified in MIL-STD-1364 may be used provided the command or agency concerned approves a written request submitted by the contractor.

4.1.2.2 Test performance levels. Procedures and techniques, developed to perform all types of maintenance tests for each specified level of 3.4.8.4, shall be verified prior to submission of (1) final data of MIL-STD-1345, (2) Navy Training Instructions and (3) Final Navy Technical Manuals. Preliminary procedures, pamphlets, or Manuals, not necessarily in the required technical format, or preliminary manuals in the standard format, may be used to verify support maintenance test procedures.

4.1.2.3 Test procedure verification. The support maintenance test procedure verification shall be performed on the first article by a technician selected by the Government from contractor or Government personnel using the following procedure:

(a) The technician shall be given a complete set of boards and modules which are completely unaligned.
(b) The technician shall align each of the modules or boards and verify that each board or module is operating properly. Tests will be done with the modules or boards on the bench.
(c) The technician shall install the modules or board in the prime equipment and align the prime equipment as specified in the manual.
(d) The technician shall operate the equipment as specified in the manual, and verify that the equipment meets specification requirements, such as output power, modulation, spurious emissions, and so forth.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing shall be classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)
(3) Environmental inspection (see 4.4.3)

(c) Reliability testing (see 4.4.4)
(d) Maintainability demonstration (see 4.4.5)
(e) Inspection of preparation for delivery (see 4.4.7)

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4.3 First article inspection. Unless otherwise specified (see 6.2) one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the inspection specified in table I.

Table I. First article inspection.

Examination or test	Requirement paragraph	Test method paragraph
Quality Conformance Inspection		4.4
Production Inspection		4.4.1
Production Control Inspection		4.4.2
Environmental Inspection		4.4.3
Visual		
Parts Selection	3.2	
Nonstandard parts	3.2.1	
Accessibility	3.5.2.3.2	
External connections	3.5.2.3.6	
Circuit test points	3.4.6.2.1	
Shielded input	3.5.1.4.4	
Output impedance	3.5.1.5.1	
RF output connector	3.5.2.2	
Equipment cooling	3.5.2.3.5	

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the requirements of table II.

Table II. Production inspection.

Examination or test	Requirement paragraph
Operational requirements	3.4.1
Burn-in	3.2.4.1
Test equipment, built-in	3.4.6.2.2
Metefs	3.4.6.2.3
Elapsed time meter	3.4.6.2.3.1
Labeling	3.4.6.2.4
Alarm lamps	3.4.7.2.2
Auditory alarms	3.4.7.2.3
Transmission cycle	3.5.1.2

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Table II. Production inspection (Continued).

Examination or test	Requirement paragraph
Pulses per frequency	3.5.1.3
Pulse repetition frequency	3.5.1.3.1
Pulse characteristics	3.5.1.3.2
Sub-pulse shape	3.5.1.3.2.1
Time and frequency accuracy	3.5.1.4.1
Stability	3.5.1.4.2
Transmitter programmer	3.5.1.4.3
Battery power automatic operation and features	3.5.1.4.5
Power output	3.5.1.5.2
VSWR matching capabilities	3.5.1.5.3
Output power metering	3.5.1.5.4
Manual operation	3.5.1.5.9
Filament voltage regulation	3.5.1.5.11
Side tracks positioning and cable lengths	3.5.2.3.3, 3.5.2.3.4
Interlocks	3.5.2.3.7

4.4.1.1 Transmitter power output. To demonstrate compliance with 3.5.1.5.2 measurements shall be made at each and every one of the following sounding frequencies taken from 3.1.1. The power output shall be measured using a 50 ohm (non-reactive) dummy load. Variation, from the 30 kw requirement shall not be greater than -1 dB or +2 dB.

(a)	2.025	2.575	3.025	3.525	3.975
(b)	4.050	5.050	6.050	7.050	7.950
(c)	8.100	10.100	12.100	14.100	15.90
(d)	16.20	20.20	24.20	28.20	31.80

4.4.1.1.1 The above frequencies shall be used to test the transmitter while operating into each of the following load impedance resulting in a VSWR of 3:1. All values below are ±5 percent. Variations in the following conditions in power output to the load shall not exceed the limits specified in 3.5.1.5.3.

	R	jX
(1)	150	0
(2)	17	0
(3)	30	-40
(4)	30	+40

Tests will be performed at the impedance values of 85 - j68 and 85 + j68 at selected frequencies corresponding to the test load requirements.

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level

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shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table III. Production control inspection.

Examination or test	Requirement paragraph	AQL percent defective
Primary power source	3.4.1.1	6.5
Step start	3.4.1.1.1	6.5
Safety	3.4.7.3.4	6.5
Spurious emission	3.4.9.3	6.5
Stability check	3.5.1.4.2, 3.5.1.4.2.1	6.5
High VSWR and Crowbar protection circuits	3.5.1.5.5	6.5
Power output control	3.5.1.5.8	6.5
Enclosures	3.5.2.3, 3.5.2.3.1	6.5
Heat test	MIL-E-16400	6.5

4.4.2.1 High VSWR circuits. It shall be demonstrated that the VSWR protection circuitry will function to remove the transmitter from operation on the frequencies of 4.4.1.1 before any damage occurs. An adequate test of this circuit will consist of operating the transmitter into each of the following impedances which result in values of 3:1, 4:1, and 5:1. It shall be shown that the protection circuit can be adjusted to operate at each of these values of VSWR. All values below are ±5 percent.

VSWR 3:1		VSWR 4:1		VSWR 5:1	
R	jX	R	jX	R	jX
150	0	200	0	250	0
17	0	13	0	10	0
30	-40	24	-44	20	-46
30	+40	+44	+44	20	+46

Tests shall be performed at VSWR of 3:1 at impedance values of 85 - j68 and 85 + j68, and at VSWR of 4:1 with impedance values of 100 - j93 and 100 + j93, and at VSWR of 5:1 with impedance values of 120 - j120 and 120 + j120. These tests shall be performed at frequencies compatible with the test loads.

4.4.2.2 Crowbar protection. It shall be demonstrated that the electronic Crowbar protection circuitry is capable of sensing incipient overvoltage and overcurrent conditions and removing the transmitter from operation before any component damage occurs. The values of overvoltage and overcurrent necessary to operate the electronic Crowbar circuit will be dependent on the voltage and current ratings and dissipation capabilities of the tubes and other components in the transmitter. An adequate test of this circuit will consist of the

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simulation of these values of voltage and current and confirmation that the transmitter does remove itself from operation without damage.

4.4.2.3 Spurious emissions. The spurious and harmonic radiation produced by each of the transmitted frequencies of 4.4.1.1 shall be measured in accordance with the key-down limits for test CE06, outside the emissions bandwidth, as specified in 3.4.9.3. For these tests, a calibrated spectrum analyzer may be used while the transmitter is operating at full power into a 50 ohm dummy load.

4.4.2.4 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to, and passed, production control inspections may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all visible damage.

4.4.3 Environmental inspection. Environmental inspection shall be as listed in table IV and shall be performed on units which have been subjected to and passed production inspection. The thermal survey may be excluded after having been performed once unless changes to equipment design or fabrication occur which may affect thermal characteristics.

Table IV. Environmental inspection.

Examination or test	Requirement paragraph
Temperature	3.3, 3.4.2, MIL-E-16400
Humidity	3.4.2, MIL-E-16400
RF field power	3.4.2
Elevation	3.4.2
Shock and vibration	3.4.3
Electromagnetic characteristics	3.4.9
Stability check	3.5.1.4.2.1
Accelerated life test	MIL-E-16400
Thermal survey	3.4.4

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments produced or fraction thereof. The first sample equipment shall be selected from the first months production.

4.4.3.1.1 Humidity tests. The equipment shall be subjected to the Range 4 humidity test cycles specified in MIL-E-16400, except the filament and blower circuits shall be energized two hours prior to completion of the last cycle.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same

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materials, processes, and so forth. If the Quality Assurance Representative does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.4).

4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to, and passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.4.3.4 Thermal survey. To verify proper thermal design of the equipment, the contractor shall conduct a complete thermal survey of the AN/FPT-11 during high temperature environmental testing, to determine the existence of Hot Spots (see 6.5) which may prove to be injurious to the useful life of the equipment. The survey shall be conducted with the equipment packaged and enclosed in the proposed final production configuration. The survey shall include sufficient temperature measurements to provide a thermal profile of the internal configuration of the equipment and all critical items at both 25°C ambient and the maximum specified ambient operating temperature.

4.4.3.5 Electromagnetic compatibility test. Compliance with 3.4.9 shall be verified by tests performed in accordance with MIL-STD-462.

4.4.4 Reliability demonstration. The requirement of 3.4.5 shall be demonstrated on the first production units in accordance with test plan VI, test level A-1 of MIL-STD-781. The demonstration shall be successfully completed prior to delivery of any production equipments.

4.4.4.1 Reliability test procedure. Throughout reliability testing, the equipment shall be operating in the 160 frequency stepping mode at full rated output power. As a minimum, the following characteristics shall be monitored daily to verify satisfactory equipment operation:

(a) Transmission cycle, initiation time and pulses per frequency (3.5.1.2 and 3.5.1.3).
(b) Pulse characteristics (3.5.1.3.2 and 3.5.1.3.2.1).
(c) Time and frequency accuracy (3.5.1.4.1).
(d) Phase stability (3.5.1.4.2).
(e) Power output (3.5.1.5.2).
(f) VSWR and Crowbar protection circuits (3.5.1.5.6).
(g) Built-in test functions (3.4.6.2.2).

4.4.4.2 Failure. The final decision as to the relevancy or nonrelevancy of a failure shall be made by the procuring activity using the Failure Categories paragraph of MIL-STD-781 as a guide. Any operating discrepancy or degradation that requires an unscheduled adjustment or calibration to be made, after initial satisfactory operation of the equipment under test, shall be defined as a relevant failure.

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4.4.4.3 Equipment burn-in. All equipment shall be subjected to a burn-in at ambient temperature while operating in the worst case (maximum total power) mode for a minimum cumulative operating time of 96 hours. Failures occuring during the first 72 hours shall be repaired and the burn-in resumed. Failures occuring during the final 24 hours shall be repaired and the equipment burned in for a minimum of 24 additional failure free operating hours prior to acceptance. Equipment monitoring during burn-in shall include as a minimum the functional checks required to assure proper operation under the worst case operating mode. No adjustments (for example, manual compensation for frequency drifts, and so forth) other then normal operating mode changes shall be allowed during equipment burn-in.

4.4.5 Maintainability demonstration test.

4.4.5.1 Maintainability demonstration. Conformance to the maintainability requirements shall be demonstrated in accordance with method 3 of MIL-STD-471. The maintainability demonstration shall be successfully completed prior to the delivery of any production equipment. The maintainability test shall be performed by a technician with experience equivalent to that of a third class Naval Electronics Technician (ET-3). The test shall include provisions to verify Part I of the prediction, that is, both module and part replacement. The maintainability requirements shall be demonstrated by replacement of modules and replaceable parts at the organizational level. Technical demonstration to be used shall be limited to the equipment technical manual. If the technical manual is not available, a preliminary draft shall be substituted.

4.4.6 Human engineering design verification test. A human engineering verification shall be performed to test the equipment and system of this specification in accordance with the requirements of the Failure Analysis paragraph and the Human Engineering Design Verification - Test and Evaluation paragraph of MIL-H-46855 (see 6.3). Tests to be included in the verification testing are:

(a) Checklists
(b) Critical operation and maintenance scenarios including normal and degraded operational modes
(c) Maintainability demonstration monitoring
(d) Technical Manual validation monitoring
(e) Selected system tests (that is, operating tests, noise level measurements, speech intelligibility tests, and so forth).

4.4.7 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by suppliers shall be specified in the individual order.)

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5.1 Preservation, packaging, packing, and marking. The preparation for delivery requirements to be called for in the procurement documents shall be in accordance with the applicable levels specified in MIL-E-17555 and MIL-STD-129.

6. NOTES

6.1 Intended use. The equipment covered in this specification is intended for Naval Shore use in conjunction with sounder receiving equipment aboard Naval ships and at shore communications stations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Marking, levels of preservation and packaging and packing required (see 5.1).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Environmental inspection. Approval to ship deliverable items may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.5 Definitions.

6.5.1 Military off-the-shelf equipment. Equipment for which the engineering design and production drawings have been finalized and the first article testing has been accomplished and accepted.

6.5.2 Commercial off-the-shelf equipment. Equipment for which the engineering design and production drawings have been finalized.

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6.5.3 Hot spot. Any item (part or assembly of parts) used in the equipment including, but not limited to, semiconductor devices, printed circuit boards, transformers, resistors, and so forth, which attain a temperature in excess of that recommended by the item manufacturer or applicable military specification, or an assembly of parts that attain a temperature which could cause accelerated degradation of the equipment and system, or both.

Preparing activity:
Navy - EC

(Project 5840-N172)

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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